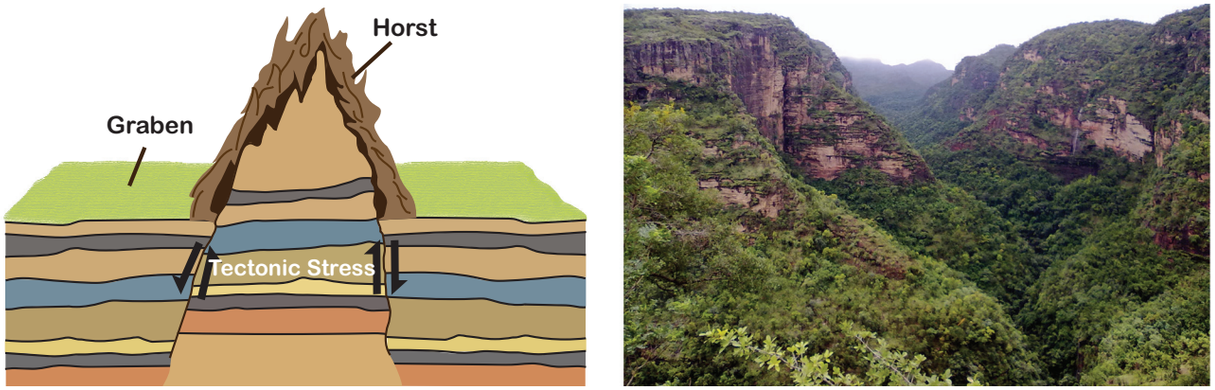


Figure 3.20 Horst (Left) and Vindhya Block Mountain (Right)

of Africa, Dead Sea of Israel and the Red Sea form the parts of the Great Rift Valley.

2. Horst

When a block of land between two faults is pushed up, block mountain or horst is formed. In this case, the central block is not only up thrown but the side blocks are also relatively downthrown so that the whole central mass appears like a dome.

In India, specifically the mountain ranges of Vindhya and Satpura found in the central western part of the India are block mountains.

3.8.1.2 Reverse Fault

A reverse fault is a horizontal displacement of the crust. It is formed where two fractured blocks move towards each other. It is caused by compressional forces along convergent plate boundaries. One side of the fault lies at an angle above the other.

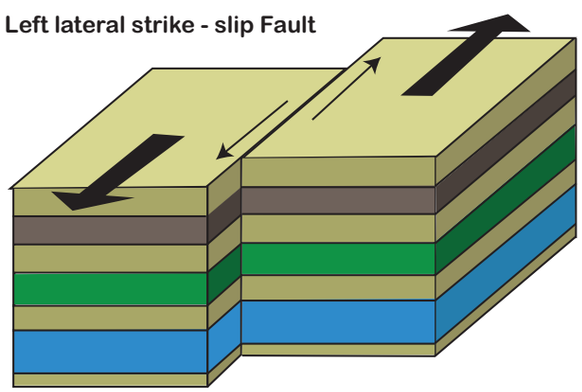


Figure 3.21 Transform fault

3.8.1.3 Shear Fault / Transform Fault/ Strike – Slip Fault

It is created by shearing along transform boundaries. Rocks on either side of fault slip past each other sideways with little up or down motion. It mostly occurs in the ocean basin and connects offsets in the mid ocean ridge.

3.9 Earthquake

Earthquake is a sudden shaking of the earth's surface. **Focus** is the location inside the earth where the earthquake originates. **Epicenter** is the point on the earth's surface vertically above the focus of an earthquake. Earthquake results from the sudden release of pressure which has slowly built up within the earth's crust. Energy is released in the form of shockwaves known as seismic waves. The seismic waves can broadly be classified into two types namely Body waves and surface waves.

I. ***Body Waves*** are the waves that travel through the interior of the earth. They are further divided into the following.

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